

AREHI-SECOPS

ASEAN Regional E-Commerce Hybrid Infrastructure & Hardened SecOps System

Phase 1 — Regional On-Premises Network

Full-Depth Configuration & Certification Study Guide

Architect / Author: Danial Nur Irfan bin Azeze

Bachelor of Computer Science — Networks & Security, UTM

CompTIA Security+ Certified

Exam objectives referenced: Cisco CCNA 200-301, CompTIA Security+ SY0-701, AWS Certified Solutions

Architect – Associate (SAA-C03)

Lab platform: Cisco Packet Tracer 9.0.0

Confirmed lab device models:

MY-KL-HQ-CORE — Catalyst 3650-24PS

SG-EDGE-GW — ISR 4331

PH-MNL-ROAS / TH-BKK-ROAS — ISR 4321

PH-MNL-ACC / TH-BKK-ACC — Catalyst 2960-24TT

Contents

1. Project & Topology Overview
2. A Note on Exam Objective Tags
3. Objective 1 — Basic Device Identity & Access Control
4. Objective 2 — VLAN Database
5. Objective 3 — LAN-Side IP Addressing
6. Objective 4 — Spanning Tree
7. Objective 5 — LAN Trunking to Neighbor Device
8. Objective 6 — WAN Links
9. Objective 7 — Cloud-Facing Interface
10. Objective 8 — Access Ports (End-Host Facing)
11. Objective 9 — Discovery Protocols
12. Objective 10 — Management Access (SSH)
13. Platform Lessons Learned (Packet Tracer Realities)
14. Real Bugs Found During Verification
15. Appendix — Quick Reference Tables

1. Project & Topology Overview

This guide documents Phase 1 of the AREHI-SECOPS project: building the ASEAN regional on-premises network for a cross-border K-pop e-commerce business headquartered in Malaysia, with regional fulfillment hubs in the Philippines and Thailand, and a cloud on-ramp in Singapore. It is written to explain not just *what* was typed, but *why*, and to define every networking-specific term used along the way — at the depth expected for real exam understanding, not just lab completion.

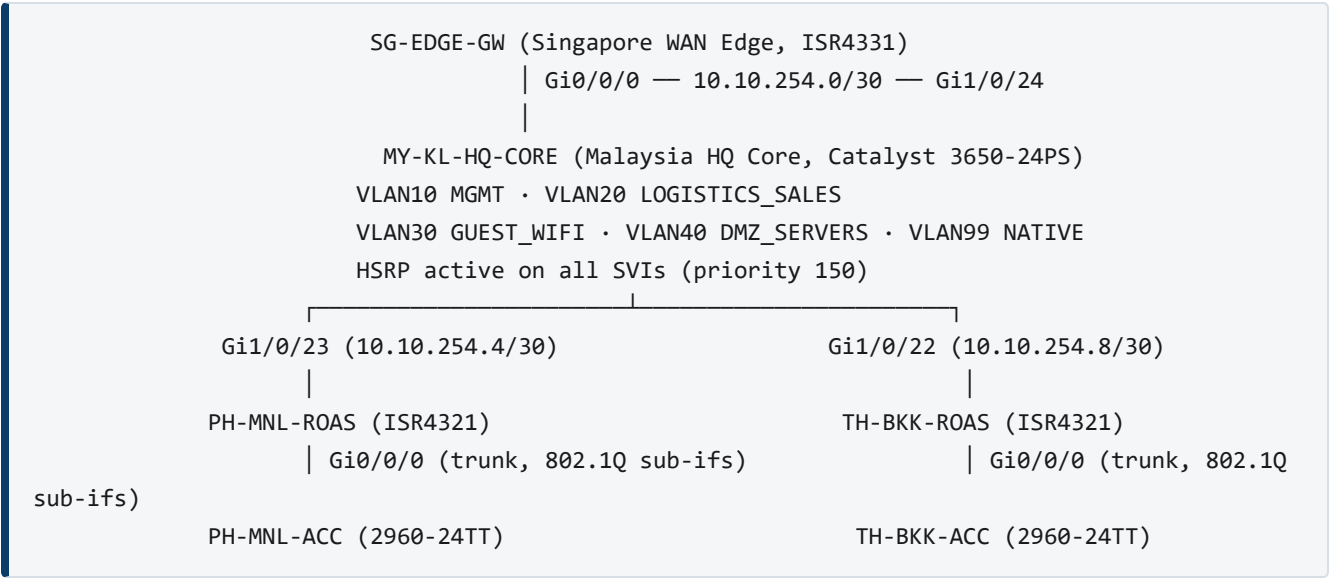
Device inventory (confirmed Packet Tracer models)

Hostname	Role	Confirmed PT model	Site
MY-KL-HQ-CORE	Core/distribution switch (Layer 3)	Catalyst 3650-24PS	Kuala Lumpur, Malaysia
SG-EDGE-GW	WAN edge / cloud on-ramp router	ISR 4331	Singapore
PH-MNL-ROAS	Router-on-a-stick	ISR 4321	Manila, Philippines
PH-MNL-ACC	Access switch (Layer 2)	Catalyst 2960-24TT	Manila, Philippines
TH-BKK-ROAS	Router-on-a-stick	ISR 4321	Bangkok, Thailand
TH-BKK-ACC	Access switch (Layer 2)	Catalyst 2960-24TT	Bangkok, Thailand

WHY THE CONFIRMED MODELS MATTER

The `.cfg` reference files in this repository intentionally still describe an aspirational real-world deployment target (e.g. Catalyst 9300 for the core) — that's the platform a genuine production rollout would use. The table above is what this specific Packet Tracer lab was actually built and verified on, which is why certain commands behave differently than the files' header comments imply (documented throughout this guide, and consolidated in Section 13).

Topology (hub-and-spoke, Malaysia HQ as hub)



IPv4 VLSM addressing plan (full reference — each objective below repeats only the slice relevant to it)

Site	VLAN	Name	Subnet	Gateway	Subnet mask / prefix
Malaysia	10	MGMT	10.10.10.0/24	.1 (HSRP VIP)	255.255.255.0
Malaysia	20	LOGISTICS_SALES	10.10.20.0/24	.1 (HSRP VIP)	255.255.255.0
Malaysia	30	GUEST_WIFI	10.10.30.0/24	.1 (HSRP VIP)	255.255.255.0
Malaysia	40	DMZ_SERVERS	10.10.40.0/24	.1 (HSRP VIP)	255.255.255.0
Manila	10	MGMT	10.10.110.0/24	.1 on PH-MNL-ROAS	255.255.255.0
Manila	20	LOGISTICS_SALES	10.10.111.0/24	.1 on PH-MNL-ROAS	255.255.255.0
Manila	30	GUEST_WIFI	10.10.112.0/24	.1 on PH-MNL-ROAS	255.255.255.0
Manila	40	DMZ_SERVERS	10.10.113.0/24	.1 on PH-MNL-ROAS	255.255.255.0
Bangkok	10	MGMT	10.10.120.0/24	.1 on TH-BKK-ROAS	255.255.255.0
Bangkok	20	LOGISTICS_SALES	10.10.121.0/24	.1 on TH-BKK-ROAS	255.255.255.0
Bangkok	30	GUEST_WIFI	10.10.122.0/24	.1 on TH-BKK-ROAS	255.255.255.0
Bangkok	40	DMZ_SERVERS	10.10.123.0/24	.1 on TH-BKK-ROAS	255.255.255.0
WAN transit	—	MY↔SG	10.10.254.0/30	MY=.1, SG=.2	255.255.255.252
WAN transit	—	MY↔Manila	10.10.254.4/30	MY=.5, PH=.6	255.255.255.252
WAN transit	—	MY↔Bangkok	10.10.254.8/30	MY=.9, TH=.10	255.255.255.252

WHY THESE SPECIFIC MASKS — VLSM LOGIC

Every LAN subnet uses /24 (255.255.255.0, 254 usable hosts) — generous for a VLAN that might grow, and it makes the third octet double as a readable site/VLAN identifier (e.g. .20.x = LOGISTICS_SALES everywhere). The WAN links use /30 (255.255.255.252) instead — a point-to-point link only ever has exactly 2 devices on it, and a /30 gives exactly 2 usable host addresses (network + broadcast take the other 2 of the 4 total addresses in the block). Using a /24 on a link that only ever needs 2 addresses would waste 252 addresses per link for no reason — this contrast is exactly what **Variable Length Subnet Masking (VLSM)** means: choosing a different mask length per subnet based on how many hosts that specific subnet actually needs, instead of using one uniform mask everywhere.

2. A Note on Exam Objective Tags

Every objective below is tagged with whichever of the three certifications it genuinely maps to — no forcing all three onto everything. Phase 1 is almost entirely on-premises Cisco networking, so **CCNA 200-301** tags appear on nearly every objective. **Security+ SY0-701** tags appear wherever a hardening/security control is the actual point of the objective (port security, ACL-adjacent concepts, BPDU/Root Guard, disabling discovery protocols on untrusted links, SSH). **AWS SAA-C03** appears exactly *once* in this phase (Objective 7) — Phase 1 is the on-premises phase of this project; the cloud integration work happens in Phase 4, so most of Phase 1 legitimately has no AWS objective to map to, and this guide won't pretend otherwise.

3. Objective 1 — Basic Device Identity & Access Control

All 6 devices

CCNA 200-301: Domain 5.0 (Security Fundamentals)

Security+ SY0-701: Domain 2.0 (Architecture) & 4.0 (Security Operations)

Purpose: before any traffic-forwarding feature is touched, every device needs an identity and a locked-down way to manage it. This objective establishes who the device is and who is allowed to configure it — the foundation everything else sits on. Skipping this and configuring VLANs/routing first would mean running an unmanaged, unauthenticated device on a live network, even briefly.

Commands, word by word

```
hostname MY-KL-HQ-CORE
enable secret P@ssw0rd
service password-encryption
no ip domain-lookup
ip domain-name danny-sec.internal
username asean.admin privilege 15 secret Adminp@ss
```

Word / term	Meaning
hostname	Sets the device's name — used in the CLI prompt, in neighbor-discovery advertisements (CDP/LLDP), and in every log message and SIEM entry this device ever generates. Not just cosmetic — it's how you identify which device did what, later.
enable	Short for "enable mode" — Cisco's term for privileged EXEC mode, the access level where configuration changes are possible (as opposed to user EXEC mode, which only allows read-only commands like <code>show</code> and <code>ping</code>).
secret	Specifically means the password is stored as an irreversible cryptographic hash (MD5-based type 5, or SCRYPT-based type 9 depending on IOS version) — as opposed to the older <code>enable password</code> command, which stores the password in a reversible, much weaker form. "Secret" vs "password" is a real, tested distinction.
service password-encryption	A global command that weakly (reversibly — type 7, not cryptographically secure) obscures any <i>other</i> plaintext passwords sitting in the configuration, such as old-style line passwords. It does not affect <code>enable secret</code> , which is already properly hashed regardless.
domain-lookup	IOS's built-in behavior of trying to resolve any unrecognized CLI input as a hostname via DNS. <code>no ip domain-lookup</code> turns this off — without it, a mistyped command hangs for up to 30 seconds while the device fruitlessly tries to resolve your typo as a DNS name.
ip domain-name	Sets this device's DNS domain suffix. This isn't just informational — it's a hard prerequisite for RSA key generation (Objective 10), because the generated key is literally named <code>hostname.domain-name</code> . No domain name means no SSH key means no SSH.
username	Creates an entry in the device's local user database, used for authentication when no external server (like RADIUS/TACACS+) is configured.
privilege 15	Cisco IOS has privilege levels 0–15. Level 1 is the default for a new user — user EXEC mode, read-only. Level 15 is full privileged EXEC — the same access level as knowing the <code>enable secret</code> . Omitting this defaults a new user to level 1, meaning they could log in via SSH but couldn't type <code>enable</code> or make any configuration change.

EXPECTED RESULT

```
show running-config | include username
```

This should print the username line back, including `privilege 15` and a hashed (never plaintext) secret. If `privilege 15` is missing after you're sure you typed it, the most common cause is re-typing the `username` line a second time without the privilege keyword — re-declaring the same username overwrites the entire previous entry, silently resetting privilege back to the default of 1.

PLATFORM REALITY CONFIRMED IN THIS LAB

On the Catalyst 2960-24TT access switches specifically, `privilege 15` was found to not persist in `show running-config` even when typed correctly — confirmed as a platform quirk on that model, not a configuration mistake, and deliberately not chased further since it doesn't block the rest of the project's goals.

4. Objective 2 — VLAN Database

MY-KL-HQ-CORE, PH-MNL-ACC, TH-BKK-ACC (switches only — routers have no VLAN database)

CCNA 200-301: Domain 2.0 (Network Access)

Security+ SY0-701: Domain 3.0 (Security Architecture — network segmentation)

Purpose: a **VLAN (Virtual Local Area Network)** is a way to logically split one physical switch into multiple separate broadcast domains, as if each VLAN were on its own dedicated switch hardware, without buying separate switches. This objective only declares that the VLANs *exist* and names them — no IP addressing, no port assignment yet, those come later.

The VLAN scheme and why each one is separated

VLAN	Name	Purpose	Why it's isolated from the others
10	MGMT	Device management traffic (SSH, future WLC/AP management)	Admin access must never share a broadcast domain with regular user or guest traffic
20	LOGISTICS_SALES	Finance and order-fulfillment staff — the actual production business traffic	The most business-critical VLAN — gets the strictest port security in Objective 8
30	GUEST_WIFI	Wireless guest access	Deliberately isolated from MGMT, DMZ, and LOGISTICS_SALES — a guest device should never be able to reach internal business systems
40	DMZ_SERVERS	TFTP, SIEM, RADIUS — semi-trusted infrastructure services	Reachable by legitimate internal traffic but kept separate from end-user segments
99	NATIVE	Catch-all for untagged trunk traffic	Deliberately carries no real traffic at all — a classic hardening move against VLAN-hopping attacks that specifically abuse whichever VLAN is configured as native

Commands, word by word

```
vlan 10
name MGMT
```

Word / term	Meaning
vlan 10	Creates VLAN ID 10 in the switch's VLAN database and enters VLAN configuration submode for it. VLAN IDs 1–1005 are the classic range; 1002–1005 are reserved by the switch itself for legacy FDDI/Token Ring and always exist by default.
name	Assigns a human-readable label to the VLAN ID — purely for administrative clarity, has no effect on switching behavior. <code>MGMT</code> and <code>10</code> refer to the exact same VLAN either way.

EXPECTED RESULT

```
show vlan brief
show vtp status
```

Why `show running-config` won't show this: on VTP-backed platforms (confirmed true for the 3650-24PS and 2960-24TT used here), VLAN definitions live in a separate database file (`vlan.dat`), not in the running-config text — `show vtp status` reporting VTP Operating Mode: Server confirms this is why. A quick sanity check on the count: every switch ships with 5 default VLANs (1, plus 1002–1005). If `show vtp status` reports 10 existing VLANs after adding the 5 custom ones (10/20/30/40/99), the math confirms all 5 were created successfully ($10 - 5 = 5$).

REAL BUG FOUND DURING VERIFICATION

VLAN 20's name was independently typed as `LOGISTIC_SALES` (missing the "S") on more than one live device — caught via `show vlan brief`, fixed by re-entering `vlan 20 / name LOGISTICS_SALES` (re-declaring a VLAN name overwrites the old one, no delete needed first).

5. Objective 3 — LAN-Side IP Addressing

MY-KL-HQ-CORE, PH-MNL-ACC, TH-BKK-ACC, PH-MNL-ROAS, TH-BKK-ROAS

CCNA 200-301: Domain 2.0 (Network Access) & Domain 3.0 (IP Connectivity)

Purpose: give each VLAN an actual Layer 3 presence — a gateway address end devices can route through. The mechanism differs by device role, which is itself an important distinction: a device either has the hardware/software capability to route between VLANs directly (a Layer 3 / multilayer switch), or it doesn't and needs an external router's help (router-on-a-stick).

3a. MY-KL-HQ-CORE — SVIs with HSRP (true inter-VLAN routing)

```
ip routing
!
interface Vlan10
  description MGMT - Core switch / AP / WLC management
  ip address 10.10.10.2 255.255.255.0
  standby version 2
  standby 10 ip 10.10.10.1
  standby 10 priority 150
  standby 10 preempt
  no shutdown
```

Word / term	Meaning
ip routing	Critical, easy to forget: a Cisco multilayer switch does NOT route between VLANs by default — unlike a router, which routes by default. Without this global command, SVIs still get IP addresses and HSRP still negotiates, but the switch never actually forwards a single packet between VLANs.
SVI (Switch Virtual Interface)	A virtual (not physical) interface representing a VLAN's Layer 3 identity on a multilayer switch. <code>interface Vlan10</code> creates the SVI for VLAN 10 — every device physically switched into VLAN 10 uses this SVI's IP as its gateway.
standby	The keyword for HSRP (Hot Standby Router Protocol) , a Cisco-proprietary First Hop Redundancy Protocol (FHRP). "Standby" is HSRP's own terminology for the whole feature — the command doesn't just mean "backup," it's literally the protocol name.
standby version 2	Selects HSRPv2 over the older v1 — v2 supports VLAN/group numbers above 255 and uses a faster default hello timer.
standby 10 ip 10.10.10.1	The <code>10</code> here is the HSRP <i>group number</i> (commonly matched to the VLAN ID for clarity, but technically independent of it) — and <code>10.10.10.1</code> is the virtual IP (VIP) , the address end devices actually use as their default gateway. This is deliberately different from the switch's own physical SVI address (<code>10.10.10.2</code>).
priority 150	This switch's bid to become the active router for this HSRP group — default priority is 100, higher wins the election. With only one switch in this lab, it wins trivially; this value matters the moment a second HQ switch is added as a standby peer.
preempt	Allows this switch to automatically reclaim active status if it recovers after a failure, rather than leaving a lower-priority standby permanently in charge once it has taken over.

EXAM ANGLE — FHRP CONCEPT

HSRP protects against the failure of *the device your traffic first exits through* — not just a link failure. The virtual IP is not tied to one physical device; if a second switch existed as standby and the active one failed, the standby would take over the VIP within seconds, completely transparently to every end device on that VLAN. VRRP (an open-standard alternative) and GLBP (Cisco's load-balancing variant) are the other two FHRPs CCNA expects you to be able to distinguish from HSRP.

3b. PH-MNL-ACC / TH-BKK-ACC — management-only SVI (not routing)

```
interface Vlan10
 ip address 10.10.110.2 255.255.255.0
 no shutdown
!
 ip default-gateway 10.10.110.1
```

Word / term	Meaning
<code>ip default-gateway</code>	Different command from <code>ip route</code> — this is the correct way for a Layer-2-only switch (one with <code>ip routing</code> never enabled) to have an off-subnet path for its own management traffic. It only affects traffic <i>originated by the switch itself</i> , not traffic passing through it (a pure L2 switch doesn't route passing traffic at all).

KEY DISTINCTION FROM 3A

These switches are Layer 2 only — `ip routing` is never enabled here. This SVI exists purely so the switch itself can be managed/reached over the network; it does not route between VLANs. Real traffic between VLANs at this site is handled entirely by the router in 3c below.

3c. PH-MNL-ROAS / TH-BKK-ROAS — router-on-a-stick sub-interfaces

```
interface GigabitEthernet0/0/0
  no shutdown
!
interface GigabitEthernet0/0/0.10
  encapsulation dot1Q 10
  ip address 10.10.110.1 255.255.255.0
!
interface GigabitEthernet0/0/0.99
  encapsulation dot1Q 99 native
```

Word / term	Meaning
Sub-interface (<code>Gi0/0/0.10</code>)	A logical subdivision of one physical interface — the router treats <code>Gi0/0/0.10</code> , <code>.20</code> , <code>.30</code> , <code>.40</code> as four separate interfaces, each capable of its own IP address and its own routing behavior, all riding over the one physical cable.
<code>encapsulation</code>	Specifies what tagging/framing format this sub-interface expects incoming frames to use, so the router knows how to interpret and strip the VLAN tag before further processing. Without an encapsulation statement, a sub-interface exists but never actually processes any tagged traffic — this is a functional requirement, not a decoration.
<code>dot1Q</code>	Short for IEEE 802.1Q , the industry-standard VLAN tagging protocol — a 4-byte tag inserted into an Ethernet frame that identifies which VLAN the frame belongs to. This is the only encapsulation type in real-world use today; Cisco's older proprietary alternative (ISL) has been obsolete for years, which is also why some modern switch platforms in this lab no longer even expose an encapsulation choice (see Section 13).
<code>encapsulation dot1Q 99 native</code>	The <code>native</code> keyword marks VLAN 99 as this trunk's native VLAN — the one VLAN whose frames are sent <i>untagged</i> . Correctly has no IP address configured, since native VLAN 99 deliberately carries no real routable traffic in this design.

EXAM ANGLE — WHY ROUTER-ON-A-STICK EXISTS AT ALL

Manila and Bangkok's access switches (2960-24TT) are Layer 2 only, so they cannot host SVIs to route between VLANs themselves. Router-on-a-stick solves this with a single physical link carrying all VLANs (tagged via 802.1Q) to an external router, which performs the routing on the switch's behalf via sub-interfaces — this is the direct real-world alternative whenever the switch hardware itself lacks Layer 3 capability.

IP addressing reference for this objective

Device	Interface	IP address	Role
MY-KL-HQ-CORE	Vlan10 / Vlan20 / Vlan30 / Vlan40	10.10.10.2 / .20.2 / .30.2 / .40.2 (physical) + .1 HSRP VIP on each	Real Layer 3 routing via SVIs
PH-MNL-ACC	Vlan10	10.10.110.2	Management only, default-gateway → .110.1
PH-MNL-ROAS	Gi0/0/0.10 / .20 / .30 / .40	10.10.110.1 / .111.1 / .112.1 / .113.1	Real routing gateway for all 4 Manila VLANs
TH-BKK-ACC	Vlan10	10.10.120.2	Management only, default-gateway → .120.1
TH-BKK-ROAS	Gi0/0/0.10 / .20 / .30 / .40	10.10.120.1 / .121.1 / .122.1 / .123.1	Real routing gateway for all 4 Bangkok VLANs

EXPECTED RESULT

```
show ip interface brief
show standby brief    (MY-KL-HQ-CORE only)
```

All configured interfaces should show `up / up`, and `show standby brief` should list all four HSRP groups as `Active` with the correct virtual IP.

REAL BUG FOUND — THE MOST SERIOUS ONE IN PHASE 1

On `PH-MNL-ROAS`, the `.10` sub-interface was found completely missing both `encapsulation dot1q 10` and its IP address — likely skipped by accident while typing. This meant `10.10.110.1` did not exist anywhere on the network, silently breaking the switch's off-subnet reachability, despite the physical link showing fully up. This is a textbook illustration of a fault that looks fine at Layer 1/2 (cable up, port up) but is broken at Layer 3 — caught only by methodically checking `show ip interface brief` on every sub-interface individually, not just the parent.

6. Objective 4 — Spanning Tree

MY-KL-HQ-CORE, PH-MNL-ACC, TH-BKK-ACC

CCNA 200-301: Domain 2.0 (Network Access)

Security+ SY0-701: Domain 4.0 (Security Operations — Layer 2 hardening)

Purpose: prevent Layer 2 loops. Ethernet frames have no TTL or hop-count mechanism the way IP packets do — a frame with an unknown destination gets flooded out every port, and a physical loop makes that flood repeat forever, doubling with every pass until the network melts down (a broadcast storm). Spanning Tree elects one root bridge per VLAN and blocks any redundant path back to it, while keeping the blocked path ready to activate instantly if the primary fails.

Commands, word by word

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,99 root primary
spanning-tree portfast bpduguard default
```


Word / term	Meaning
<code>rapid-pvst</code>	Rapid PVST+ — Cisco's implementation of 802.1w (Rapid Spanning Tree Protocol) , combined with PVST+ (Per-VLAN Spanning Tree Plus) , meaning one independent spanning-tree instance runs <i>per VLAN</i> rather than one shared instance for the whole switch. "Rapid" refers to convergence speed: after a topology change, RSTP converges in roughly 1–2 seconds versus 30–50 seconds for the original 802.1D standard.
<code>root primary</code>	A macro command — IOS automatically lowers this switch's bridge priority below whatever it currently observes from other switches on the network, guaranteeing it wins the root election. Resolves in <code>show running-config</code> to a literal fixed value, <code>priority 24576</code> .
<code>root secondary</code> (used on PH-MNL-ACC)	Same idea, but sets a fixed backup priority (<code>28672</code>) — a deliberately chosen, predictable fallback root, rather than leaving the election to whichever switch happens to have the lowest MAC address by accident.
<code>portfast</code>	Skips the normal Listening → Learning delay (30 seconds total) on ports connected to end hosts, since a PC doesn't need to wait through loop-detection states the way a switch-to-switch link does. Meant only for access ports, never trunk links to other switches.
<code>bpduguard</code>	BPDU (Bridge Protocol Data Unit) Guard — a BPDUs are the actual messages switches exchange to run Spanning Tree. If a port marked PortFast (assumed to be an end-host connection) ever receives a BPDUs, that means a switch — not a PC — was plugged in, and BPDUs Guard immediately error-disables the port. A real, concrete defense against someone plugging in a rogue switch and accidentally or deliberately creating a loop.

EXAM ANGLE — ROOT ROLE DESIGN CHOICES IN THIS TOPOLOGY

MY-KL-HQ-CORE (the hub) is root primary — the root bridge should always be the most central, reliable device, since every spanning-tree path calculation radiates outward from it. PH-MNL-ACC is root secondary — a predictable, chosen backup. TH-BKK-ACC deliberately gets no explicit root role — it still fully participates in the protocol, it's just never a candidate to become root itself, which is realistic: a regional warehouse switch two WAN hops from HQ isn't a sensible fallback root in a real network.

EXPECTED RESULT

```
show spanning-tree summary
show spanning-tree vlan 10
```

Confirms the reported Root ID matches the expected switch's own bridge ID. One caveat worth knowing: since these switches connect to each other only via routed WAN links through their ROAS routers (not a direct Layer 2 trunk between switches), each switch is Spanning-Tree-isolated from the others — every one will report itself as root for its own VLANs, simply because it has no Layer 2 neighbor to lose an election against. That's expected in this topology, not a misconfiguration.

PLATFORM REALITY CONFIRMED IN THIS LAB

`spanning-tree ?` on the 3650-24PS and 2960-24TT only exposes `mode` , `portfast` , and `vlan` — no `extend` or `loopguard` keyword exists. Further testing revealed `spanning-tree extend system-id` is actually a mandatory, always-on feature on this platform (confirmed present in a live device's own `show run` despite no way to type it) — not missing, just non-configurable. `loopguard default` , by contrast, was confirmed genuinely absent via direct CLI rejection (`% Invalid input detected`). Full detail in Section 13.

7. Objective 5 — LAN Trunking to Neighbor Device

MY-KL-HQ-CORE, PH-MNL-ACC, TH-BKK-ACC

CCNA 200-301: Domain 2.0 (Network Access)

Security+ SY0-701: Domain 4.0 (Security Operations — DTP hardening)

Purpose: the physical link between each switch and the next device down the chain, carrying multiple VLANs tagged with 802.1Q over a single cable — as opposed to an access port, which only ever carries one VLAN, untagged.

Commands, word by word

```
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,99
switchport nonegotiate
channel-protocol lacp
channel-group 1 mode active
```

Word / term	Meaning
<code>switchport mode trunk</code>	Explicitly forces the port into trunking mode, rather than access mode or letting negotiation decide.
<code>trunk native vlan 99</code>	Sets which VLAN's frames cross this trunk untagged — deliberately the unused VLAN 99, so there's nothing meaningful to exploit via a native-VLAN-hopping attack (a technique where an attacker crafts double-tagged frames to jump into a VLAN they shouldn't have access to).
<code>trunk allowed vlan</code>	Explicitly restricts which VLANs may cross this specific trunk. Without this, a trunk allows <i>every</i> VLAN by default, including ones with no legitimate reason to be on this particular link.
<code>nonegotiate</code>	Disables DTP (Dynamic Trunking Protocol) — the Cisco protocol that lets two connected switches automatically negotiate whether a link becomes a trunk. Hard-coding trunk mode on both ends and disabling negotiation closes a known attack: a rogue device can otherwise pretend to be a switch, negotiate a trunk via DTP, and gain access to every VLAN on it.
<code>channel-protocol lacp</code>	LACP (Link Aggregation Control Protocol) , the IEEE 802.3ad standard for bundling multiple physical links into one logical one — chosen here over Cisco's older proprietary alternative, PAgP.
<code>channel-group 1 mode active</code>	Assigns this physical interface to EtherChannel bundle number 1. <code>active</code> means this side actively initiates LACP negotiation, rather than passively waiting to be asked (<code>passive</code> mode) — at least one side of any LACP bundle must be active for negotiation to happen at all.

EXAM ANGLE — ETHERCHANNEL AND SPANNING TREE TOGETHER

Bundling two physical links (EtherChannel) gives both extra bandwidth *and* redundancy — but critically, Spanning Tree treats the entire bundle as one single logical link, so it never blocks one of the two members as "redundant" the way it would with two separate, unbundled links. You get the bandwidth and the loop protection simultaneously, which is exactly why EtherChannel is generally preferred over leaving two parallel links unbundled.

EXAM ANGLE — A REAL HARDWARE CONSTRAINT THAT CHANGED THIS DESIGN MID-PROJECT

The ISR 4321 routers used at Manila and Bangkok only have **2 onboard Gigabit ports**. A 2-member LACP bundle to the access switch would consume both, leaving nothing for the WAN link back to HQ — so those two sites use a single trunk link instead of EtherChannel, freeing the second port for WAN. Only `MY-KL-HQ-CORE` (with more available ports) uses a real EtherChannel in this design.

EXPECTED RESULT

```
show interfaces trunk
show etherchannel summary    (MY-KL-HQ-CORE only)
```

Confirms the trunk is up, carrying the correct native and allowed VLANs.

PLATFORM REALITY CONFIRMED IN THIS LAB

`switchport trunk encapsulation dot1q` does not exist as a command at all on the 3650-24PS or 2960-24TT — `switchport trunk ?` only lists `allowed` and `native`. This matches real modern fixed-configuration Catalyst hardware, which removed ISL trunking entirely since only 802.1Q is ever supported — there is no alternative encapsulation left to choose, so the command was removed rather than left as a pointless single-option prompt.

8. Objective 6 — WAN Links

MY-KL-HQ-CORE, SG-EDGE-GW, PH-MNL-ROAS, TH-BKK-ROAS

CCNA 200-301: Domain 1.0 (Network Fundamentals — subnetting/VLSM) & Domain 3.0 (IP Connectivity)

Purpose: the routed hub-and-spoke links tying each remote site back to Malaysia HQ. This is pure Layer 3 addressing readiness — dynamic routing (OSPF) is deliberately deferred to Phase 2, so for now these are directly-connected point-to-point links plus a static default route on each spoke as a temporary stopgap.

Commands, word by word

```
interface GigabitEthernet1/0/23
  no switchport
  ip address 10.10.254.5 255.255.255.252
  no shutdown
!
ip route 0.0.0.0 0.0.0.0 10.10.254.5
```

Word / term	Meaning
no switchport	Converts what would normally be a Layer 2 switch port into a routed port with its own IP address — required on a multilayer switch to dedicate a physical interface to pure IP routing instead of Layer 2 switching.
ip route 0.0.0.0 0.0.0.0 <next-hop>	A static default route — 0.0.0.0 0.0.0.0 is the special "match anything" network/wildcard-mask pair, meaning "if no more specific route exists, send it here instead." This is explicitly a Phase 1 stopgap; Phase 2 replaces it with a dynamically-learned OSPF route, which is why it isn't the final design.

WAN transit addressing reference

Link	Subnet	MY side	Remote side
MY ↔ Singapore	10.10.254.0/30	.1 (Gi1/0/24)	.2 (SG-EDGE-GW Gi0/0/0)
MY ↔ Manila	10.10.254.4/30	.5 (Gi1/0/23)	.6 (PH-MNL-ROAS Gi0/0/1)
MY ↔ Bangkok	10.10.254.8/30	.9 (Gi1/0/22)	.10 (TH-BKK-ROAS Gi0/0/1)

EXPECTED RESULT

```
ping 10.10.254.6    (from MY-KL-HQ-CORE, to PH-MNL-ROAS)
show ip interface brief
```

Every WAN link should ping its directly-connected neighbor successfully. Cross-site LAN reachability (e.g. Manila VLAN20 to Malaysia HQ) will **not** work yet — MY-KL-HQ-CORE has no route back into the spoke sites' internal VLANs, and the spokes only have a default route toward HQ. This is expected, not a bug, until Phase 2's OSPF replaces the static routes entirely.

9. Objective 7 — Cloud-Facing Interface

SG-EDGE-GW only

CCNA 200-301: Domain 3.0 (IP Connectivity)

AWS SAA-C03: Design Secure Architectures (Site-to-Site VPN / Virtual Private Gateway concepts)

Purpose: prepare the interface that will eventually source the IPsec Site-to-Site VPN tunnel to AWS in Phase 4. At this stage it only gets a placeholder public IP and a default route — no real internet/AWS device exists in the topology yet. This is the one and only place in Phase 1 with a direct, genuine tie to the AWS SAA-C03 exam, since it's explicitly the on-premises half of a future Site-to-Site VPN connection to an AWS Virtual Private Gateway — a concept SAA-C03 tests directly.

Commands, word by word

```
interface GigabitEthernet0/0/1
  ip address 203.0.113.2 255.255.255.248
  no shutdown
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

Word / term	Meaning
203.0.113.0/24	Defined by RFC 5737 as a reserved documentation-only range — along with 192.0.2.0/24 and 198.51.100.0/24, this block is guaranteed to never be assigned as a real, routable public address. Used here specifically so this placeholder can never accidentally conflict with genuine internet addressing.
255.255.255.248 (/29)	An 8-address block (6 usable) — sized generously enough for this placeholder ISP-facing segment without being as wasteful as a full /24.

EXPECTED RESULT

```
show ip interface brief
```

Gi0/0/1 should show administratively up, but with no active line protocol — nothing is physically cabled to it yet, since there's no real internet/AWS device in this lab. That's correct for this stage, not an error; it changes once Phase 4 builds the AWS side.

10. Objective 8 — Access Ports (End-Host Facing)

MY-KL-HQ-CORE, PH-MNL-ACC, TH-BKK-ACC

CCNA 200-301: Domain 2.0 (Network Access) & Domain 5.0 (Security Fundamentals)

Security+ SY0-701: Domain 4.0 (Security Operations — endpoint/port hardening)

Purpose: harden the ports that connect real end devices — staff workstations, barcode scanners, and guest wireless AP uplinks — against Layer 2 attacks such as MAC flooding and unauthorized device connections.

Commands, word by word

```
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree guard root
```

Word / term	Meaning
port-security	Enables the feature on this port at all — every sub-command below only takes effect once this base command is present.
maximum 2	Caps how many distinct source MAC addresses this port will learn/allow simultaneously. A number higher than 1 here typically accounts for something like a PC plus a VoIP phone daisy-chained through it.
sticky	Instead of the administrator manually typing each allowed MAC address, the switch dynamically learns the first MAC(s) it sees on the port and automatically writes them into the running-config as if they'd been statically configured.
violation restrict	Defines what happens when a MAC beyond the configured maximum is detected: traffic from the offending MAC is dropped and logged, but the port itself stays up and continues passing traffic from already-allowed MACs. This is less severe than the default violation mode, <code>shutdown</code> , which error-disables the entire port on any violation.
spanning-tree guard root	Prevents this port from ever becoming a path toward a <i>better</i> (lower-priority) root bridge. If the port receives a BPDU claiming superiority, indicating an unexpected switch pretending to be a better root, the port is blocked instead of letting Spanning Tree recalculate the topology around it.

EXAM ANGLE — THE STANDARD "HARDENED ACCESS EDGE" PATTERN

Port security, BPDU Guard, and Root Guard together form a pattern tested across both CCNA and Security+: assume every access port could have something malicious plugged into it, and constrain what that specific port is allowed to do, regardless of what shows up on it.

EXPECTED RESULT

```
show port-security interface FastEthernet0/1  
show vlan brief
```

Confirms port security shows enabled with the correct maximum/sticky/restrict settings, and that `show vlan brief` now lists real ports assigned under VLAN 20/30, rather than everything still sitting on the default VLAN.

11. Objective 9 — Discovery Protocols

All 6 devices

CCNA 200-301: Domain 1.0 (Network Fundamentals)

Security+ SY0-701: Domain 4.0 (Security Operations — information leakage on untrusted links)

Purpose: CDP and LLDP broadcast this device's identity, platform, and sometimes IP/VLAN information to whatever's directly connected — extremely useful for troubleshooting, but information that should never leak out toward a WAN edge or any externally-facing link, since it hands a potential attacker a head-start on reconnaissance.

Commands, word by word

```
cdp run
lldp run
interface range GigabitEthernet1/0/22 - 24
  no cdp enable
  no lldp transmit
  no lldp receive
```

Word / term	Meaning
CDP (Cisco Discovery Protocol)	Cisco's proprietary Layer 2 neighbor-discovery protocol — works only between Cisco devices.
LLDP (Link Layer Discovery Protocol)	The vendor-neutral IEEE 802.1AB equivalent of CDP — works between any vendor's equipment that supports it, which is why both are configured here rather than relying on CDP alone.
no cdp enable	Disables CDP on this specific interface only, without turning off the globally-enabled protocol everywhere else.
no lldp transmit / no lldp receive	LLDP's per-interface disable is split into two separate directions — this device stops both advertising itself outward and listening for/accepting advertisements inward, on this interface.

EXPECTED RESULT

```
show cdp neighbors
```

LAN-side links (e.g. an access switch to its ROAS router) should show the neighbor. WAN-facing links (any MY-KL-HQ-CORE spoke link, SG-EDGE-GW 's interfaces) should show **nothing**, even though the link itself is fully up and pinging fine — that's the suppression working as intended.

12. Objective 10 — Management Access (SSH)

All 6 devices

CCNA 200-301: Domain 5.0 (Security Fundamentals)

Security+ SY0-701: Domain 2.0 (Architecture — secure remote access, encryption in transit)

Purpose: replace any unencrypted management access with SSH, so credentials and configuration changes are never sent across the network in cleartext.

Commands, word by word

```
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
line vty 0 15
  transport input ssh
  login local
```

Word / term	Meaning
crypto key generate rsa	Generates an RSA public/private keypair — the asymmetric cryptography SSH uses to establish a secure session before any password is ever exchanged. This is an <i>action</i> command, not a stored configuration line — it never appears in <code>show running-config</code> even when it worked correctly, the same way flipping a physical switch isn't something you'd expect to see written down afterward.
general- keys	Generates a single general-purpose keypair used for both signing and encryption, as opposed to <code>usage-keys</code> , which generates two separate keypairs (one for each purpose) — the simpler, more common choice.
modulus 2048	The RSA key length in bits — 2048 is the current practical minimum for real security; anything shorter (512/768/1024) is considered crackable with modern computing power and shouldn't be used.
ip ssh version 2	Forces SSHv2 only — SSHv1 has known cryptographic weaknesses and should never be enabled on a real device.
line vty 0 15	VTY (Virtual Teletype) lines are the virtual terminal sessions used for remote access (Telnet/SSH) — <code>0 15</code> configures all 16 of them (numbered 0 through 15) identically in one pass.
transport input ssh	Restricts these lines to SSH only — no unencrypted Telnet permitted, even though IOS supports it by default otherwise.
login local	Authenticates incoming connections against the local username database (the account from Objective 1), rather than a single shared line password with no per-user accountability.

EXAM ANGLE — A PREREQUISITE CHAIN WORTH NOTICING

This objective only works because of decisions made all the way back in Objective 1: `ip domain-name` had to be set before RSA keys could generate, and the `username ... privilege 15` account had to exist before `login local` meant anything. Exam questions frequently test exactly this kind of "what has to be configured first" dependency chain, not just the final command in isolation.

EXPECTED RESULT

```
show crypto key mypubkey rsa
show ip ssh
ssh -l <username> <target-ip>  (from another device's privileged EXEC prompt)
```

With no PCs cabled into this lab, SSH is tested device-to-device: from one router's CLI, SSH into a directly reachable neighbor's management IP. Since OSPF isn't in place yet (Phase 2), this only works between devices with a direct IP path — directly-connected WAN neighbors, or devices on the same LAN subnet.

13. Platform Lessons Learned (Packet Tracer Realities)

Observation	Resolution
<code>spanning-tree extend system-id</code> and <code>loopguard default</code> not selectable via the interactive <code>spanning-tree ?</code> menu on the Catalyst 3650-24PS / 2960-24TT	<code>extend system-id</code> turned out to be a mandatory always-on feature (confirmed present in a live device's own running-config despite no toggle existing) — not missing, just non-configurable. <code>loopguard</code> was confirmed genuinely unsupported via direct CLI rejection.
<code>switchport trunk encapsulation dot1q</code> not accepted	Confirmed via <code>switchport trunk ?</code> showing only <code>allowed / native</code> — matches real modern fixed-configuration Catalyst hardware, which is dot1Q-only with no ISL alternative to select.
ISR 4321 routers only have 2 onboard Gigabit ports (<code>Gi0/0/0</code> , <code>Gi0/0/1</code>)	Forced a design change from a 2-member LACP EtherChannel to the access switch, down to a single trunk link — freeing the second port for the WAN uplink.
2960-24TT port naming — only 2 fixed Gigabit ports (<code>Gi0/1</code> , <code>Gi0/2</code>) plus a bank of <code>FastEthernet0/x</code> ports	Not the Catalyst 9200L stack-style <code>GigabitEthernet1/0/x</code> naming originally assumed in the reference files. All access-layer interface names were translated accordingly.
<code>username ... privilege 15</code> doesn't persist in <code>show running-config</code> specifically on the 2960-24TT	Logged as a known quirk of that specific model; doesn't affect the rest of the lab's goals, so deliberately not chased further.

THE UNIVERSAL TEST FOR "MISSING" VS "DEFAULT" COMMANDS

Type the exact command directly in configuration mode. `% Invalid input detected` means genuinely unsupported on this platform, not a display artifact. Accepted with no error, but the line still doesn't show up afterward in `show run` — that's a default/redundant command, already true, nothing wrong.

14. Real Bugs Found During Verification

Every device's live `show running-config` was audited line-by-line against the intended design after initial configuration. This caught several genuine mistakes a surface-level glance would have missed:

Device	Bug	Severity	Fix
PH-MNL-ROAS	<code>Gi0/0/0.10</code> sub-interface missing both <code>encapsulation dot1Q 10</code> and its IP address entirely	Functional — broke the switch's off-subnet reachability, though the physical link showed as fully up	Re-entered both missing lines on the sub-interface
MY-KL-HQ-CORE	VLAN 20's SVI description and the VLAN name itself both read <code>LOGISTIC_SALES</code> (missing "S")	Cosmetic	Retyped both the <code>description</code> and <code>vlan 20 / name</code> lines
PH-MNL-ACC, TH-BKK-ACC	Same VLAN name typo independently repeated on both access switches	Cosmetic	Same fix, re-entering the VLAN name
PH-MNL-ROAS	VLAN 20 sub-interface description had a typo introduced <i>while fixing</i> the first one (<code>warehous</code> instead of <code>warehouse</code>)	Cosmetic	Retyped the description again

WHY THIS SECTION MATTERS FOR EXAM PREP

The one functional bug is a perfect illustration of a classic troubleshooting trap: Layer 1 and Layer 2 can look completely healthy (cable connected, port up, trunk negotiated) while Layer 3 is silently broken underneath. Methodical, interface-by-interface verification with `show ip interface brief` — not just "the link light is on" — is what catches this class of fault.

15. Appendix — Verification Command Quick Reference

What to check	Command
VLANs actually exist	<code>show vlan brief</code> , <code>show vtp status</code>
Interface up/down state	<code>show ip interface brief</code>
Trunk health	<code>show interfaces trunk</code>
EtherChannel state	<code>show etherchannel summary</code>
Spanning-tree role/root	<code>show spanning-tree summary</code> , <code>show spanning-tree vlan <id></code>
HSRP state	<code>show standby brief</code>
Port security status	<code>show port-security interface <if></code>
CDP/LLDP neighbors	<code>show cdp neighbors</code> , <code>show lldp neighbors</code>
RSA keys / SSH readiness	<code>show crypto key mypubkey rsa</code> , <code>show ip ssh</code>
Direct connectivity test	<code>ping <ip></code> , <code>ssh -l <user> <ip></code>
Full config dump	<code>show running-config</code>